PYTHON // GUI PROGRAMMING

Building Graphical Interfaces with Tkinter

Concept walkthrough followed by a live coding demonstration in VS Code

Introduction to Python

SESSION PLAN

Agenda for Today

1

Concept Overview (10 min)

This presentation — what Tkinter is, core widgets, layout managers, and event-driven programming, with sample code for each idea.

2

Live Demo (remaining time)

Coding each example directly in VS Code and running it, so students see the window and output appear in real time.

GETTING STARTED

What is Tkinter?

Tkinter is Python's standard, built-in library for building graphical user interfaces (GUIs) — no separate installation required.

1. **Window**
2. **Label**
3. **Button**
4. **Entry box**
5. **Text box**
6. **Checkbutton**
7. **Radio button**
8. **Listbox**
9. **Menu**

```
Basic structure:

import tkinter
window = tkinter.Tk()
window.mainloop()
```

mainloop() keeps the window running and waits for user actions (clicks, typing, etc.)

CORE FLOW

How a Tkinter Program Runs



Once mainloop() starts, the window stays open and reacts to events (button clicks, keyboard input) until it is closed.

```
import tkinter as tk

window = tk.Tk()
window.title("My First GUI")
window.geometry("400x300")
window.mainloop()
```

tk.Tk() creates the main window
window.title(...) sets the title
window.geometry(...) sets width x height
window.mainloop() starts the event loop

DISPLAY & ACTION WIDGETS

Label and Button

Label

Displays text or information on the window.

```
label = tk.Label(
    window,
    text="Welcome to Python GUI"
)
label.pack()
```

Button

Performs an action when clicked, via command=.

```
def say_hello():
    print("Hello!")

btn = tk.Button(
    window, text="Click Me",
    command=say_hello)
```

Click flow:



TAKING INPUT

Entry Widget & Reading User Input

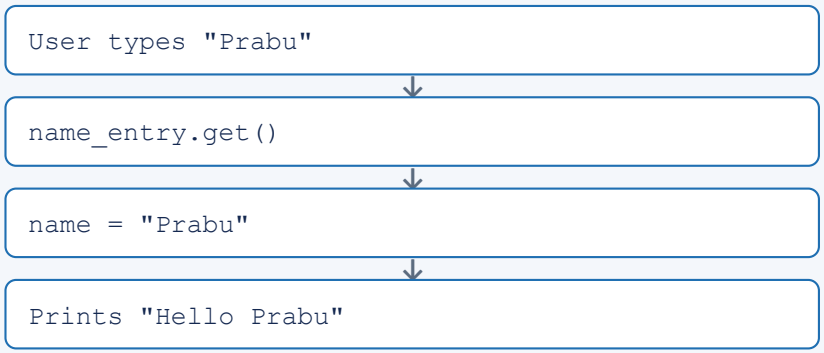
Entry lets the user type text. The get() method reads what was typed.

```
def show_name():
    name = name_entry.get()
    print("Hello", name)

name_entry = tk.Entry(window)
name_entry.pack()

btn = tk.Button(
    window, text="Submit",
    command=show_name)
btn.pack()
```

Flow



REFLECTING OUTPUT ON SCREEN

Updating a Label Instead of Printing

Instead of printing the result in the terminal, we change a Label's text so the user sees the result inside the GUI itself.

```
def show_name():
    name = name_entry.get()
    result_label.config(
        text="Hello " + name
    )

result_label = tk.Label(
    window, text=""
)
result_label.pack()
```

Before click

Result label: (empty)

After click (name = Prabu)

Result label: Hello Prabu

ARRANGING WIDGETS

Layout Managers: pack, grid, place

1 pack()

Simple top-to-bottom stacking. Easiest for beginners.

```
label.pack()
button.pack()
```

2 grid()

Rows & columns — ideal for form-style layouts.

```
label1.grid(row=0, column=0)
entry1.grid(row=0, column=1)
```

3 place()

Exact x, y coordinates. Precise but less flexible.

```
button.place(x=100, y=150)
```

For beginners, pack() and grid() cover almost every layout need.

A DIFFERENT WAY OF THINKING

Event-Driven Programming

Normal Program

Statement 1

↓

Statement 2

↓

Statement 3

↓

End

GUI Program

GUI Window → Wait for Event

Button Click | **Keyboard Input** | **Mouse Click**

`Function()` | `Function()` | `Function()`

Common events you'll use: Button click, Mouse click, Keyboard input, Mouse movement, Window closing.

ALERTS & POPUPS

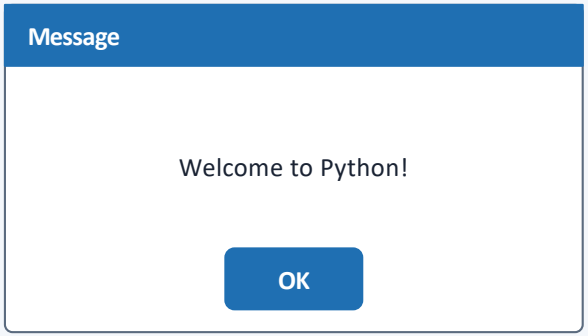
Message Box

Tkinter can show popup messages using the messagebox module — useful for confirmations, warnings, and simple alerts.

```
from tkinter import messagebox

def show_message():
    messagebox.showinfo(
        "Message",
        "Welcome to Python!"
    )

btn = tk.Button(
    window, text="Click Me",
    command=show_message
)
```



PUTTING IT TOGETHER

Mini Project: Student Greeting App

A single program that combines every concept covered so far —
an ideal capstone example before moving to independent practice.




```
def greet_student():
    name = name_entry.get()
    if name == "":
        messagebox.showwarning(
            "Warning",
            "Please enter your name"
        )
    else:
        result.config(
            text="Welcome, " + name + "!"
        )
```

SUMMARY

Core Widgets & Concepts to Teach First

Widgets

- `Tk()`
- `Entry`
- `Text`
- `Radiobutton`
- `messagebox`
- `Label`
- `Button`
- `Checkbutton`
- `Listbox`
- `Frame`

Concepts

- Creating a window
- Adding widgets
- Layout: pack() / grid()
- Getting user input
- Button events
- Functions as handlers
- Updating widgets
- Message boxes
- Input validation
- Event-driven programming

Next: Live coding demo in VS Code — one .py file per concept, run and observe the output together.